# From Human–Human Collaboration to Robot Assistance in Collaborative Assembly

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***Abstract*— For many assembly tasks, effective human–robot collaboration can benefit from both physical assistance and context-aware interaction. However, which robotic capabilities should be incorporated remains an open question. We investigate whether human–human (H2H) collaboration can provide empirical evidence for these design decisions. We analyze an existing gearbox coassembly dataset comprising 851 assistance events and 656 mentions of descriptors used to identify parts, including shapes, sizes, and spatial relations. The analysis motivates three design requirements for the studied task: support for part/tool access and workpiece stabilization, awareness of assembly progress, and flexible, context-grounded understanding of part references. We realize these requirements in H2H2R, a human–robot coassembly system that integrates robot-assisted retrieval, handover, and workholding with AR-mediated, progress-aware interaction. We evaluate the resulting capabilities through three experiments. First, a part-reference benchmark shows that LLMs can reliably identify the intended gearbox components from free-form language descriptions. Second, a study examines how participants reposition a gearbox baseboard held by the robot during assembly. Combining physical guidance with AR-based control reduces hand travel compared to physical guidance and active interaction time relative to AR-based control. Third, a part-acquisition study evaluates task-aware guidance that uses the current assembly step to narrow the candidate parts. This approach reduces workload relative to spatial selection while enabling context-dependent part requests that do not require users to fully specify each target. Together, these findings demonstrate how observed H2H behaviors can guide the design and integration of complementary robot-assistance mechanisms for shared physical tasks.**

## I. Introduction

Human–robot collaboration in shared tasks can benefit from combining physical robot assistance with interfaces that support bidirectional communication and allow users to maintain awareness of the workspace and task progress. Yet the ability of collaborative robots to effectively coordinate with humans on these tasks remains only partially realized [1, 2]. A central design challenge is deciding which assistance capabilities and interaction mechanisms should be combined, and how they should be coordinated as the shared task evolves. Collaborative assembly provides a useful testbed for this challenge because it requires both physical interaction and situated decision-making, while allowing flexibility in how people perform the task [3, 4]. Meeting these demands requires coordinating human adaptability and dexterity with robots' ability to retrieve, hold, and position components with precision and consistency [5, 6, 7].

Realizing this complementarity requires humans and robots to maintain a shared understanding of the workspace, task progress, and each other's current and intended actions.

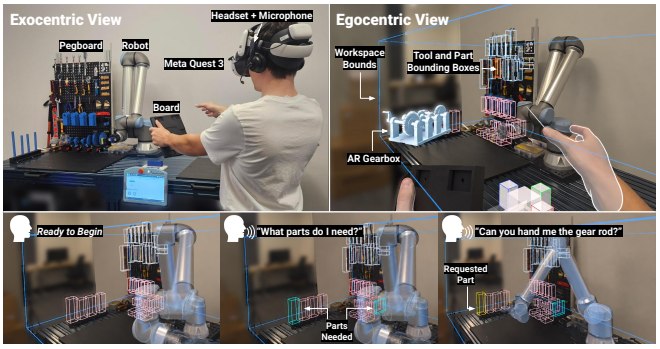



Fig. 1. **Top**: An exocentric view shows the user wearing a Meta Quest 3 headset and a microphone, together with the robot, assembly board, and pegboard containing parts and tools. The egocentric AR view shows the workspace boundary, part and tool bounding boxes, and the AR Gearbox. **Bottom**: Progress-aware assistance leverages the active assembly step to answer questions about required components (center, highlighted in cyan) and resolve part requests (right, the resolved target highlighted in yellow).

Gaze, speech, gesture, and combinations of these modalities allow users to convey their intent by indicating target objects and expressing requests for robot assistance [8, 9]. More recently, multimodal foundation models have enabled a more flexible interpretation of natural expressions grounded in workspace and task context [7, 10]. The robot can respond to these requests through physical assistance, such as component retrieval followed by handover [11] or robot-assisted workholding, where it grasps, positions, and stabilizes a workpiece during assembly [12, 13]. Beyond expressing requests, users also need feedback about robot actions and task progress. Augmented reality (AR) can support both user input and feedback by capturing spatial interactions and overlaying robot, object, and task information within the shared workspace [14, 15, 16].

Together, these technologies provide building blocks for human–robot (H2R) coassembly, but their availability alone does not determine which assistance capabilities and interaction mechanisms are needed, or when they should be used. Observations of human–human (H2H) collaboration provide a natural source of empirical evidence for these design decisions [4], revealing when assistance arises, what contextual information helpers use, and how partners communicate about shared objects and task progress. We therefore ask: *How can human–human collaboration inform which assistance capabilities and interaction mechanisms a robot should combine for a shared physical task?*

We propose **H2H2R**, an approach that uses observations of H2H coassembly to derive design requirements for H2R coassembly and realizes them in an integrated human–robot gearbox assembly system. We analyze an existing H2H

gearbox coassembly dataset collected from 20 participants interacting with a trained helper [4], examining both assistance events and the language used to identify assembly components. These observations inform three complementary capabilities: physical robot assistance, progress-aware assistance grounded in assembly state, and flexible part-reference understanding supported through AR-mediated interaction. We implement these capabilities in a physical robot system for gearbox coassembly, as illustrated in Fig. 1, and evaluate the corresponding design choices through experiments on free-form part-reference understanding, robot-assisted workholding, and task-aware part acquisition.

The contributions of this work are:

- *Empirical H2H-to-H2R design:* We translate observed assistance and part-reference patterns from H2H coassembly into three design requirements: physical assistance, progress-aware assistance, and context-grounded part-reference understanding.
- *H2H2R system:* We integrate these requirements into H2H2R, a human–robot coassembly system that combines robot-assisted manipulation with AR- and language-mediated interaction grounded in assembly progress.
- *Module-level evaluation:* We separately evaluate key H2H2R capabilities through a part-reference benchmark and two controlled user studies examining workpiece repositioning and task-aware part acquisition.

## II. Related Work

### A. From Human–Human to Human–Robot Collaboration

Human–human (H2H) collaboration has been used to derive communication and coordination mechanisms for HRI. Analyses of industrial assembly pairs have yielded a robot gesture lexicon from recurring communication terms and gestures [17], and comparisons of H2H and H2R pointing in a pick-and-place task have informed bidirectional confirmation in the robot interface [18]. H2H demonstrations have also supported learned spatial and temporal coordination for collaborative assembly [19] and context-aware gesture control for robot handling [12].

Other work has transferred physical coordination and intent models from H2H data. Analyses of human dyads have informed haptic communication principles [20] and controllers for planar co-manipulation [21], while InteRACT pretrains intent prediction on H2H interactions before adapting it to H2R manipulation [22]. These studies each transfer a single communication channel, interaction model, or control policy from human collaboration. H2H2R instead systematically analyzes H2H coassembly [4] to identify which combination of assistance functions a shared task requires, then integrates part retrieval, physical workholding, flexible part-reference understanding, and progress-aware guidance within one system.

### B. Multimodal Interaction for Human–Robot Collaboration

Pointing, gaze, speech, and mixed reality provide complementary cues for grounding human intent in a shared workspace. Visual pointing supports bidirectional object reference in pick-and-place interaction, including designs informed by H2H communication [18]. Mixed-reality deictic gestures similarly link spatial references to objects and robot actions [23]. Gaze can reduce the amount of explicit description required: gaze–speech systems combine visual attention with spoken requests [9], while GVGNet learns gaze-directed grounding for underspecified object references [24]. In collaborative assembly, GEAR uses gaze to request robot part fetching and reports lower physical demand and effort than touchscreen selection [25]. Recent assembly systems use language and foundation models to interpret natural-language instructions and generate context-aware robot actions [7, 10]. BALI combines observed actions and language for open-ended goal inference and requests clarification when needed [26]. In contrast, H2H2R uses a language model to ground free-form descriptions to inventory components and assembly progress to resolve underspecified part requests.

## III. Human–Human Coassembly Analysis

Designing a robot collaborator for a coassembly task requires understanding the forms of assistance that arise during coassembly, the task situations that trigger them, and how collaborators request and coordinate support. To obtain this empirical grounding, we analyze an existing H2H gearbox coassembly dataset, DYAD [4], originally collected for training multimodal perception models of human behavior during collaborative assembly. Specifically, we examine how the *performer* (the participant assembling the gearbox in Fig. 3A) sought and received assistance from the *helper* (a trained collaborator present across all sessions in Fig. 3A), and translate the observed assistance patterns and communication behaviors into design requirements for human–robot coassembly.

### A. Human–Human Gearbox Coassembly Setup

DYAD involves 20 participants tasked with assembling the four-row gearbox shown on the left of Fig. 2. Each row contains two shaft holders (hereafter, *gear stands*), each fastened to the baseboard with a screw and fitted with a bearing, and one shaft (hereafter, the *gear rod*) carrying one or two gears secured by pins. Row 1 additionally includes a crank handle secured to the gear rod by another pin. The rows require different screw types and corresponding tools.

During each session, the performer sat near the baseboard, while the remaining components and tools were positioned

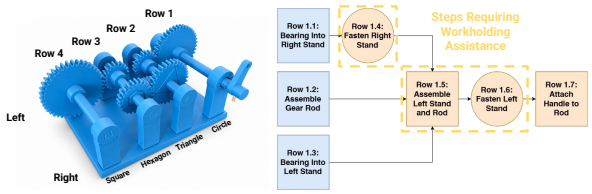



Fig. 2. **Gearbox assembly task.** Left: completed gearbox comprising four assembly rows. Right: representative task graph for Row 1, showing within-row step dependencies. Yellow dashed regions indicate steps requiring workholding assistance.

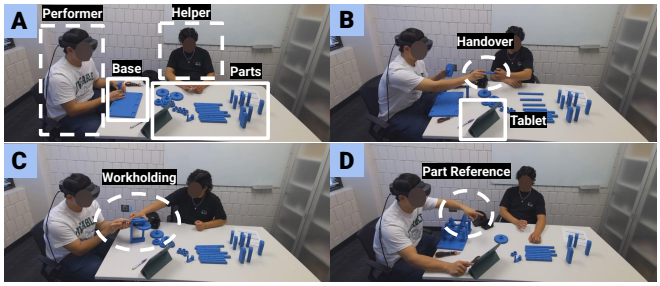



Fig. 3. **H2H coassembly setup and assistance behaviors**. (A) performer–helper setup, (B) part retrieval and handover, (C) workholding during insertion and fastening, and (D) grounding of verbal and deictic part references.

outside their direct reach (Fig. 3A). Participants inspected an interactive 3D model of the completed gearbox on a tablet (Fig. 3B) and independently determined the assembly procedure without prescribed step-by-step instructions. The four rows could be assembled in any order, but each imposed local ordering constraints that were not immediately evident. For Row 1, the two bearing insertions (Steps 1.1 and 1.3 of Fig. 2) and gear rod assembly (Step 1.2) could be completed in any order. One gear stand could then be fastened, but the gear rod must be positioned before fastening the other. Fastening both stands first blocked insertion of the gear rod and required fastening to be undone. The task therefore combines flexible progression with sequential dependencies whose violation causes assembly errors and rework.

The helper supported the performer through physical assistance and assembly guidance, including help with locating components, selecting subsequent steps, and avoiding assembly errors. Direct interruptions were generally reserved for actions likely to cause consequential errors or substantial rework. Recurring assistance behaviors included retrieving out-of-reach components requested through pointing or description (Fig. 3B), stabilizing the assembly so the performer could work two-handed for aligning and fastening (Fig. 3C), and interpreting natural-language and deictic part references within the shared workspace (Fig. 3D). The flexible assembly sequence and visually similar components create ambiguity in part selection, task progression, and task ordering, prompting both physical assistance and assembly guidance.

We conduct two complementary analyses of the H2H interactions. First, we group assistance events from DYAD according to their annotated assistance triggers to examine the forms of support that arose and the task situations that prompted them. Second, we characterize how collaborators referred to assembly components. We then translate the recurring patterns from these analyses into design requirements for H2R coassembly.

### *B. H2H Analysis and Design Implications*

The DYAD annotations comprise 851 assistance events classified using a nine-trigger schema. A blind audit of 147 events showed high annotation reliability (event-detection F1 = .990; trigger-label agreement = 84.6%, Cohen's $\kappa$ = .826) [4]. Because the same trained helper participated across sessions under a fixed guidance policy, these frequencies characterize assistance observed in this task setting rather

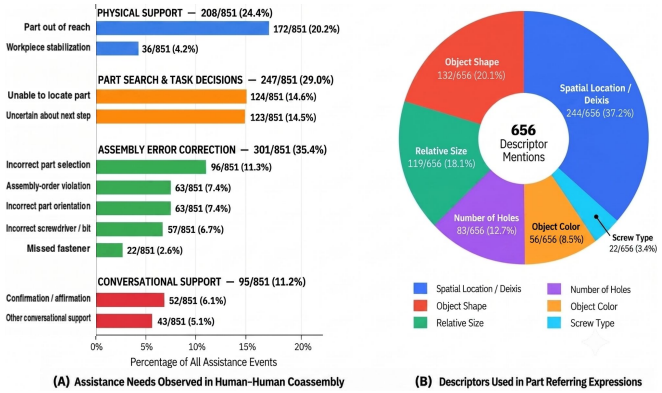



Fig. 4. **H2H coassembly analysis.** A: Distribution of assistance triggers across 851 annotated assistance events. B: Distribution of 656 descriptors used in part-referring expressions.

than universal assistance patterns. For the present design analysis, we consolidate these trigger labels into four assistance categories, as summarized in Fig. 4A. The Conversational Support category further subdivides the original *Others* trigger into confirmation/affirmation and other conversational support, yielding 11 trigger types in Fig. 4.

*1) Assistance Trigger Types:* Physical support accounted for 208 events (24.4%), primarily involving out-of-reach parts or tools (172, 20.2%) and workpiece stabilization (36, 4.2%). These observations motivate **D1: a robot collaborator should provide physical assistance for part/tool access and workpiece stabilization**. Part-search and task-decision assistance accounts for 247 events (29.0%), including cases in which performers could not locate a required component (124, 14.6%) or determine the appropriate next step (123, 14.5%). These findings motivate **D2: a robot collaborator should provide progress-aware assistance**. It should maintain explicit task and workspace context to facilitate part finding and guide assembly decisions. Error-correction assistance accounts for another 301 events (35.4%), the single largest category, including incorrect part selection, assembly-order violations, incorrect orientation, incorrect screwdriver or bit selection, and missed fasteners. H2H2R takes a preventive approach to this category through the AR Gearbox (Sec. IV-E), which lets users compare the AR and physical assembly directly to identify discrepancies before they cause errors, while active detection and correction of errors as they occur remains future work (Sec. VI).

*2) Natural Part Referring Expressions:* As shown in Fig. 4B, DYAD contains 656 descriptor mentions in participants' part references. These categories are not mutually exclusive, as a single reference could combine multiple cues, such as pointing while describing an object's shape and location. Spatial or deictic descriptors were the most common (37.2%), followed by object shape (20.1%), relative size (18.1%), number of holes (12.7%), color (8.5%), and screw type (3.4%). These frequencies indicate that participants drew on multiple cues suited to the current workspace. Spatial and deictic information predominated, while visual and categorical attributes provided additional means of distinguishing similar components. These observations motivate **D3: A robot collaborator should understand flexible,**

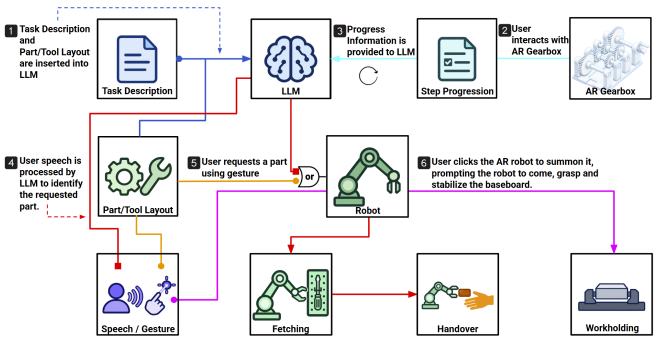



Fig. 5. **H2H2R information and interaction flow.** The part and tool layout supports language grounding and ray-based selection, while the AR Gearbox interactions update the assembly state used for progress-aware assistance. Spoken or ray-based requests initiate robotic retrieval and handover and selecting the registered AR robot initiates workholding.

**context-grounded part references without requiring predefined component names or explicit spatial-selection commands.**

Together, **D1–D3** motivate three corresponding H2H2R capabilities: robot-assisted part and tool retrieval, handover, and workpiece holding (**D1**); progress-aware assistance (**D2**); and context-grounded understanding of free-form part references (**D3**), as described in Sec. IV.

## IV. H2H2R System

Guided by the H2H-derived design requirements in Sec. III-B, H2H2R combines physical robot assistance for part and tool retrieval, handover, and workpiece holding with progress-aware AR guidance and context-grounded understanding of free-form part references.

### A. System Overview

As shown in Fig. 1, H2H2R comprises a UR10e manipulator equipped with a Robotiq 2F-85 gripper, a pegboard containing assembly components and tools, a Meta Quest 3 headset for hand-tracked AR interaction and visualization, and headphones for spoken input and verbal feedback. The AR interface overlays the physical workspace with the AR Gearbox, workspace boundaries, a virtual robot model, and interactive bounding boxes for parts and tools. The virtual robot and bounding boxes are spatially aligned with their physical counterparts. At initialization, H2H2R loads a part and tool JSON file along with a task description for the given assembly task (Item 1 in Fig. 5). For each item, the layout specifies its canonical name, robot grasp configuration, and bounding box pose and dimensions. These names define the valid outputs for language grounding, while the spatial information generates registered bounding boxes for ray-based selection. The AR Gearbox connects user interaction to the task graph: selecting or completing an operation updates the active, completed, and valid next steps supplied to the LLM (Items 2 and 3). Users can then request items through speech (Item 4) or direct ray-based selection (Item 5), with either pathway providing an item identifier for robotic retrieval and handover. The registered virtual robot provides access to the workholding capability (Item 6).

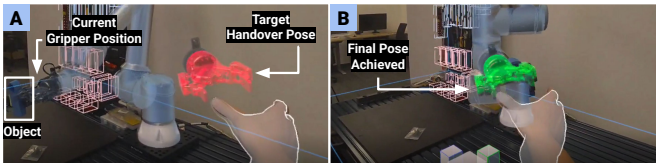



Fig. 6. **AR-guided handover.** (A) The user positions the hand-tracked ghost gripper to specify the desired handover pose. (B) Its color transitions from red to orange and then green as the robot approaches the target.

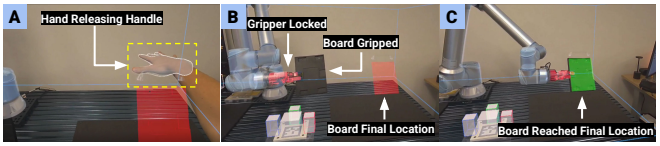



Fig. 7. **AR-guided workholding.** (A) The user releases the AR handle to confirm the target board pose. (B) The robot maintains its grasp while moving the board. (C) The robot moves the board to the target pose.

### B. Safety-Aware Robot Motion Control

To support close-proximity handover and workholding, H2H2R filters nominal operational-space velocity commands through an operational-space control-barrier-function quadratic program (OSCBF-QP) [27]. Collision spheres placed along the robot links and gripper enforce joint-position and joint-speed limits, height clearance, workspace bounds, static-obstacle avoidance (e.g., pegboard), self-collision avoidance, and dynamic clearance from the user's tracked hand. At each control step, the filter returns the feasible command closest to the nominal command, maintaining the configured geometric and kinematic constraints during execution.

### C. Part and Tool Retrieval and Handover

The retrieval and handover module addresses **D1** by bringing requested parts and tools to the user. As illustrated in Fig. 5, the user can identify an item through free-form speech (Item 4) or AR ray-based selection (Item 5). Spoken requests are grounded using the task description, current assembly state, part and tool layout, and delivery history, as detailed in Sec. IV-E. During ray-based selection, an item's AR bounding box turns orange when hovered over and green when selected. Once an item is identified, H2H2R sends its pegboard location and predefined grasp pose to the robot, which retrieves the item from the pegboard. After retrieval, the user specifies the desired handover pose by positioning the *ghost gripper*, a hand-tracked virtual model of the robot gripper. The robot then moves the item to the pose indicated by the ghost gripper. As shown in Fig. 6, the ghost gripper transitions from red to orange and then green as the robot approaches the handover pose, providing progressive feedback on its proximity to the target.

### D. Robot-Assisted Workholding

As shown in Fig. 7, robot-assisted workholding implements the stabilization behavior identified in **D1**. This capability supports Steps 4–6 of each gearbox row, illustrated for Row 1 in Fig. 2, during which the baseboard must be positioned and stabilized while the gear stands and gear rod are installed. The user initiates workholding by selecting the

registered AR robot (Item 6 of Fig. 5) using a ray or direct poke. The user then positions a ghost gripper to specify the workholding-ready pose (same as Fig. 6), and the robot moves to this target. Once the robot arrives, the user inserts the baseboard into the gripper. A Tool Center Point (TCP) force change exceeding $15\,\mathrm{N}$ triggers gripper closure, and the same threshold later triggers release once the gripper is unlocked. The grasp remains locked during kinesthetic repositioning so guidance forces are not misinterpreted as a release command. The AR gripper indicates the grasp state using green for unlocked and red for locked (Fig. 7B). With the baseboard secured, the user configures its pose through either kinesthetic or AR interaction. In kinesthetic mode (Freedrive), the user physically guides the robot-held baseboard. In AR mode, the user manipulates a virtual replica of the baseboard using an AR handle and releases the handle to confirm the target pose. The robot then moves the board to the target pose. Users switch modes by touching a virtual gripper aligned with the physical gripper. The visible AR handle indicates when AR-based control is active. To release the baseboard, the user unlocks the gripper and applies enough extraction force to exceed the threshold.

### *E. Progress-Aware AR Gearbox and Language Guidance*

To implement **D2**, H2H2R represents assembly progress using independent task graphs for each gearbox row. Each graph encodes operation prerequisites, required parts and tools, completion states, and valid next operations, allowing flexible progression while preserving local dependencies. As shown in Fig. 8, the AR Gearbox provides a spatial interface to this representation. Its global view displays the complete gearbox. Selecting a component opens a local view for its next operation, including the relevant components, an assembly animation, and controls for completing or exiting the operation. If the operation's prerequisites are unmet, the component turns red and the task state cannot advance (Fig. 8A). After an operation is completed, its component turns orange if it is needed in a subsequent operation (Fig. 8B, C) and turns green if no further interaction with it is required (Fig. 8D). The AR Gearbox can also be placed beside the ongoing physical assembly for direct comparison, helping users identify discrepancies before they cause errors. The task state also grounds language-based guidance and item requests. H2H2R can explain the active operation ("What should I do?"), recommend valid next steps ("What is the next step?"), and identify its required parts and tools ("What parts do I need?"). As illustrated in Fig. 1 and Item 3 of Fig. 5, the active operation provides context for resolving ambiguous part requests that would otherwise match multiple components and for answering step-aware questions. For example, questions such as "What parts are needed for this step?" highlight the required items in cyan, helping users locate the relevant components. H2H2R can also resolve underspecified requests such as "the gear stand" or "the gear rod" without an explicit row identifier, while supporting detailed references such as "the left gear stand in Row 1" and "the gear rod for the row with triangular stands."

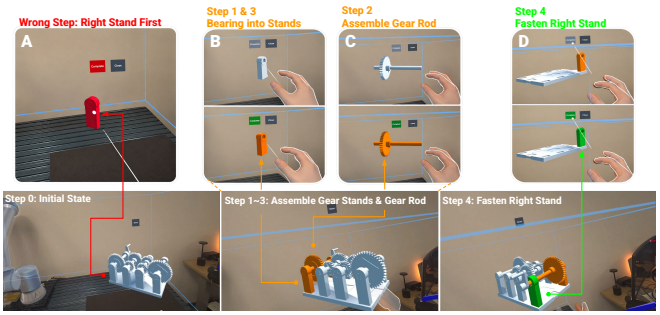



Fig. 8. **Progress-aware AR Gearbox guidance.** The global view (bottom) visualizes assembly progress and operation states. Selecting an operation opens a local view (top) with the relevant components, an animation, and step controls.

The resolved item is highlighted in yellow. The shared task graph therefore connects the progress-aware assistance of **D2** with the context-grounded language interaction of **D3**.

## V. Evaluation

To isolate module-specific effects, we conducted three experiments, each designed to evaluate a target H2H2R capability independently of the others. Experiment 1 evaluates the LLM's ability to ground free-form part-referring expressions to a fixed set of assembly items (**D3**). Experiment 2 compares kinesthetic, AR, and hybrid workholding (**D1**). Experiment 3 targets the part search effort that dominated the H2H part-search and task-decision category (29.0% of events, per Sec. III-B) by comparing ray-based (spatial interaction), free-form language, and task-aware language methods for part acquisition (**D2** and **D3**).

### *A. Participants and General Procedure*

The same 11 adults (18–31 years, $M = 24.6$, $SD = 4.2$, eight men and three women) participated in all three experiments, recruited from within the institution and compensated \$15. Mean familiarity ratings on 1–10 scales were $5.55 \pm 2.16$ for AR, $5.82 \pm 2.04$ for robotics, and $4.27 \pm 2.10$ for prior interaction with robots. The protocol was approved by the institutional IRB, and all participants provided informed consent. Experiment 1 was conducted first, familiarizing participants with the gearbox components before Experiments 2 and 3. Within Experiments 2 and 3, condition order was counterbalanced across participants.

### *B. Statistical Analysis*

In Experiments 2 and 3, subjective workload is measured using the NASA-TLX [28] (0–100 per item). All remaining constructs are measured on 7-point Likert scales anchored to construct-specific endpoints (e.g., 1 = very difficult to 7 = very easy), and multi-item questionnaire responses are averaged by construct and condition. The three conditions, including preference rankings, are compared using Friedman tests with Kendall's $W$, followed by Holm-corrected paired Wilcoxon signed-rank tests. Descriptive results are reported as mean $\pm$ sample SD, and an asterisk in the results tables (Table II, Table III) denotes $p < .05$. A sensitivity analysis indicated that $N = 11$ provides 80% power at $\alpha = .05$ for effects of $W \geq .438$. The study is therefore powered

TABLE I

EXPERIMENT 1: TOP-1 (STRICT) ACCURACY % PER MODEL

| Model | Synthetic ↑ | User-Resp. ↑ | Total ↑ | Time (s) ↓ | VRAM (GiB) ↓ |
|---|---|---|---|---|---|
| *Gemma* | | | | | |
| Gemma3-12B | **98.6** (97.2) | 95.8 (88.8) | 97.2 (93.0) | 2.65±0.30 | 25.95 |
| Gemma4-E4B | 93.8 (92.7) | 95.8 (87.1) | 94.8 (89.9) | 1.31±0.31 | 16.00 |
| Gemma3-4B | 62.8 (62.2) | 59.1 (56.6) | 61.0 (59.4) | 0.94±0.05 | 9.46 |
| *Qwen* | | | | | |
| Qwen3-VL-8B | **98.6** (94.1) | **96.2** (87.4) | **97.4** (90.8) | 1.02±0.04 | 17.75 |
| Qwen3-VL-4B | 89.2 (82.6) | 89.9 (80.4) | 89.5 (81.5) | 0.67±0.03 | 9.53 |
| Qwen3-4B | 85.4 (85.4) | 79.0 (73.8) | 82.2 (79.6) | **0.65**±0.05 | **8.75** |
| *Llama* | | | | | |
| Llama3.2-Vision-11B | 83.3 (82.6) | 73.4 (68.9) | 78.4 (75.8) | 0.92±0.04 | 21.27 |

to detect only moderate-to-large effects, and nonsignificant findings should not be interpreted as evidence of equivalence.

### C. Experiment 1: Free-Form Part Reference Understanding

*1) Design and Measures:* To evaluate **D3**, we examined whether LLMs can ground free-form part references to the correct items in the gearbox inventory. For each of the 26 gearbox components, participants viewed the assembled gearbox alongside a 3D model of the target part and wrote an unconstrained description that uniquely identified it, yielding 286 participant-generated expressions. The evaluation used these 286 expressions together with 288 synthetic expressions that we manually authored to cover technical names, visual attributes, spatial relations, and functional descriptions. We evaluated seven text and vision-language models from the Gemma [29, 30], Qwen [31, 32], and Llama [33] families. Each model received the same task description, candidate-label set, and textual referring expression, and returned one candidate label. Images were omitted to isolate language grounding and match text-only and vision-language models of similar size under identical text input. Although Qwen3-VL, Gemma3/4, and Llama-3.2-Vision support visual input, this experiment tested only their language grounding. Adding component images is left to future work. We report lenient top-1 accuracy, strict accuracy, mean inference latency, and peak GPU memory on an NVIDIA RTX 5000 Ada Generation with 32 GB of VRAM. Lenient accuracy parses a candidate label from the response, while strict accuracy requires an exact match to the target label. Models were instructed to return the group label rather than a row- or side-specific variant. Six group labels (`BEARING`, `PIN`, and `SCREW_ROW1` through `SCREW_ROW4`) each represent visually identical row- or side-specific variants. The two accuracy measures therefore differ only when a model overspecifies a valid group label, such as returning `BEARING_ROW3_LEFT` instead of `BEARING`. Incorrect parts and out-of-set outputs fail both criteria. Overspecification occurred in 132 of the 2,002 responses to participant-generated expressions (6.6%) and accounts for the entire difference between lenient and strict accuracy in Table I.

*2) Results:* Across the synthetic and participant-generated expressions, Qwen3-VL-8B achieved the highest pooled top-1 accuracy (97.4%), followed by Gemma3-12B (97.2%) and Gemma4-E4B (94.8%). On participant-generated expressions, Qwen3-VL-8B reached the highest top-1 accuracy (96.2%), while Gemma3-12B reached the highest strict accuracy (88.8%) at a higher inference cost. Qwen3-VL-4B offered a more resource-efficient alternative, reaching 89.9% accuracy at a mean latency of 0.67 s. These results show that leading open-weight models can reliably ground free-form part descriptions in this setup. We selected Qwen3-VL-8B for Experiment 3 based on this accuracy-efficiency balance.

TABLE II

EXPERIMENT 2: ROBOT WORKHOLDING

| Measure | Freedrive | AR Handle | Hybrid | $p/W$ |
|---|---|---|---|---|
| *Objective, session logs* | | | | |
| Right-hand path (m) | 2.0±0.7 | 1.5±0.2 | **1.2**±0.2 | $< .001/.84^*$ |
| Interaction time (s) | 9.3±3.5 | 9.7±4.3 | **5.8**±2.2 | $< .001/.75^*$ |
| Completion time (s) | **10.5**±4.4 | 21.0±5.8 | 15.5±3.0 | $< .001/.91^*$ |
| Interactions (#) | 2.8±1.3 | 1.8±0.3 | **1.7**±0.4 | $.041/.32^*$ |
| Trans. gain/hand (m/m) | 0.5±0.1 | 0.6±0.1 | **0.7**±0.1 | $< .001/.73^*$ |
| Rot. gain/hand (deg/m) | 48.0±12.1 | 50.8±6.5 | **61.2**±6.1 | $.002/.61^*$ |
| *Subjective, questionnaire* | | | | |
| Workload | 38.8±18.3 | 35.6±24.1 | **26.8**±16.4 | .231/.13 |
| Adj. control | 4.7±1.0 | 5.3±1.2 | **6.2**±0.7 | $< .001/.69^*$ |
| Motion predict. | **6.2**±0.7 | 4.8±1.3 | 5.7±1.1 | $.005/.48^*$ |
| Op. safety | **6.4**±1.1 | 5.6±1.1 | 6.0±0.9 | $.034/.31^*$ |
| Pref. rank | 2.6 | 2.3 | **1.2** | $.003/.52^*$ |

### D. Experiment 2: Robot Workholding

*1) Design and Measures:* To test **D1**, we used a within-participant comparison of complementary physical and virtual mechanisms for controlling the pose of a robot-held workpiece. Participants compared three interaction modes for repositioning a robot-held board.

In *Freedrive*, participants physically guided the robot-held board to the desired pose. In *AR Handle*, participants manipulated a virtual handle attached to the board to reposition a virtual representation of the board to the desired target pose. *Hybrid* allowed participants to alternate between physical guidance and AR-based manipulation during execution.

Each participant completed eight trials in every condition. Trials 1–5 emphasized translational repositioning, while Trials 6–8 required both translation and in-place rotation. A trial reached its target when position error remained below 0.05 m and rotation error remained below $15^\circ$ continuously for 1.0 s. *Adjustment Controllability* averaged the perceived simplicity of fine and coarse adjustments and the naturalness of control. *Robot Motion Predictability* combined participants' ability to anticipate the robot's movement, destination, and onset. *Perceived Operational Safety* captured safety during robot motion and confidence that the robot would not move unexpectedly. The task-specific workload composite averaged Mental Demand, Physical Demand, Effort, and Frustration. Temporal Demand was excluded because trials imposed no time constraint, and Performance was excluded because task performance was evaluated directly using the logged completion and movement metrics.

*2) Results:* The study included 11 participants, but one participant was excluded from the objective analysis because interaction and motion logs were incomplete. The resulting logs contain 240 trials from 10 participants: 80 Freedrive, 80 AR, and 80 Hybrid trials. Each trial spanned from the first interaction to the first target reach, and trial-level measures were averaged within participant and condition. Table II shows complementary benefits across the three methods.

Freedrive reached the target faster than AR and Hybrid (both $p_{\mathrm{Holm}} = .006$). Hybrid required less right-hand travel than either alternative (both $p_{\mathrm{Holm}} \leq .006$) and less interaction time (both $p_{\mathrm{Holm}} = .006$). Hybrid also achieved greater translational gain per meter of hand movement than Freedrive and AR ($p_{\mathrm{Holm}} = .006$ and .039) and greater rotational gain per meter ($p_{\mathrm{Holm}} = .008$ and .006). These results indicate that Freedrive favored rapid repositioning, while Hybrid reduced active manipulation and used hand movement more efficiently. Participants rated Hybrid as more controllable than Freedrive ($p_{\mathrm{Holm}} = .003$) and AR ($p_{\mathrm{Holm}} = .023$). They rated Freedrive and Hybrid as more predictable than AR ($p_{\mathrm{Holm}} = .035$ and .023). Operational safety differed in the omnibus test, but no corrected pairwise comparison was significant. Workload did not differ across conditions. Nine of eleven participants ranked Hybrid first, which ranked above both Freedrive ($p_{\mathrm{Holm}} = .003$) and AR ($p_{\mathrm{Holm}} = .037$) after correction. Together, these findings show a trade-off across interaction models: Freedrive enabled the fastest repositioning, while Hybrid reduced hand travel and active interaction time and received the highest controllability and preference ratings.

### *E. Experiment 3: Task-Aware Part Acquisition*

*1) Design and Measures:* To test **D2–D3**, we incrementally added language and task-state context on top of an AR spatial selection baseline. Participants requested parts under three conditions: *Gesture*, *Language*, and *Task-Aware*.

In Gesture, participants directed an AR ray at candidate objects. They identified each hovered object either through visual inspection or by hearing its spoken name through headphones, and then confirmed their selection. In Language, participants used a free-form expression that uniquely specified the desired object, such as "the gear rod for Row 2." In Task-Aware, the system constrained candidate objects using the current assembly step, task progress, and previously selected objects, letting participants use shorter context-dependent expressions such as "the gear rod," "one of them," "another one," or "the other one," interpreted relative to the parts relevant to that step.

Participants requested parts for partial assembly of Rows 1 and 2 (Fig. 2). Each condition included seven part-acquisition steps covering four gear stands, two gear rods, and one fastening task for the right stand of either row. Because background noise made speech recognition unreliable, the experimenter transcribed participants' verbal requests. The comparison therefore evaluates language grounding and task context rather than speech recognition.

*Assistance Confidence* measured participants' confidence that the system understood their requests and retrieved the correct part. *Request Ease* measured the ease of identifying and requesting a part. *Interaction Adaptation Burden* measured the extent to which participants needed to adjust their phrasing or physical position to use the method, with higher scores indicating more adjustments needed.

*2) Results:* All 11 participants completed the questionnaire.

TABLE III
EXPERIMENT 3: PART ACQUISITION

| Measure | Gesture | Language | Task-Aware | $p/W$ |
|---|---|---|---|---|
| *Objective, session logs* | | | | |
| Time/step (s) | 32.0±15.9 | 31.3±8.4 | **27.8**±7.7 | .336/.10 |
| Interactions† (#) | 274±145 | 21.1±3.2 | 21.0±1.7 | — |
| Wrong selections (%) | 19.9±9.9 | 0.0±0.0 | 0.0±0.0 | — |
| Head trans./step (m) | 1.39±1.11 | 0.67±0.45 | **0.64**±0.59 | .024/.34* |
| Head rot./step (deg) | 284.8±258.2 | 220.6±160.9 | **172.5**±128.0 | .290/.11 |
| Hand travel/step (m) | 2.86±1.08 | 1.47±1.25 | **1.13**±1.33 | .013/.48* |
| *Subjective, questionnaire* | | | | |
| NASA–TLX | 44.4±26.7 | 27.5±13.8 | **15.4**±11.4 | .010/.42* |
| Assist. conf. | 5.1±2.0 | 4.9±1.4 | **6.0**±1.1 | .214/.14 |
| Request ease | 4.1±2.0 | 5.1±1.3 | **6.2**±0.8 | .013/.39* |
| Adapt. burden | 4.1±2.1 | 3.2±1.0 | **2.8**±1.9 | .092/.22 |
| Pref. rank | 2.6 | 2.5 | **1.0** | $< .001/.75$* |

†Gesture counts hover events, whereas language conditions count transcribed request turns. These are different units and are shown only as condition-specific descriptive measures.

Gesture produced greater workload than Language ($p_{\mathrm{Holm}} = .049$) and Task-Aware ($p_{\mathrm{Holm}} = .038$). Language and Task-Aware did not differ after correction. Request Ease differed in the omnibus test, although both comparisons with Task-Aware narrowly missed significance after correction (both $p_{\mathrm{Holm}} = .053$). Assistance Confidence and Interaction Adaptation Burden did not differ across conditions. All participants ranked Task-Aware first, exceeding both alternatives after Holm correction ($p_{\mathrm{Holm}} = .003$, $W = .75$). The strong preference for Task-Aware, together with its lower workload relative to Gesture, supports the use of assembly progress to constrain part requests. Time per step and head rotation did not differ across conditions.

Task-Aware required less head translation than Gesture ($p_{\mathrm{Holm}} = .029$) and less hand travel ($p_{\mathrm{Holm}} = .035$). The other corrected pairwise movement comparisons were not significant. These differences likely stem from what each condition involved: Gesture required visually scanning the pegboard to find and point at objects, participants in Language tended to look at the AR Gearbox while composing a description, and Task-Aware's step-based narrowing made this search largely unnecessary. Gesture's hover-based selection is itself hand-driven, so its higher hand travel reflects the interaction mechanism, whereas Language and Task-Aware's values reflect only incidental movement (they had zero hover events). Gesture also had a 19.9% wrong-selection rate, consistent with participants searching by sight for the right part. In contrast, no grounding errors occurred in either language condition: across all Language and Task-Aware requests, every acquired part matched the item required for its step.

In the Task-Aware condition, task-state information contributed to interpreting 57 of 231 requests (24.7%). For example, "Which parts do I need?" highlighted the components required for the current step, while expressions such as "one of them" and "the other one" identified items among the remaining relevant candidates. The Language condition did not use task-state information and therefore did not support these context-dependent requests. Task-Aware did not significantly outperform Language overall, but it uniquely supported context-dependent requests and was unanimously preferred by participants.

## VI. Conclusion and Future Work

This work presents H2H2R, a human–robot coassembly framework that uses assistance patterns and communication behaviors observed in H2H collaboration to motivate an integrated H2R system. The H2H analysis informed three design requirements—physical assistance, progress-aware assistance, and flexible part reference—which H2H2R instantiates through robot-assisted part and tool retrieval, handover, and workholding (**D1**), progress-aware assistance (**D2**), and free-form part references (**D3**). Across three module-level experiments, our results demonstrate complementary benefits among these mechanisms. The results do not establish end-to-end superiority of the integrated system, but show how observation of H2H collaboration can inform the selection and integration of assistance mechanisms for shared tasks.

Future work will pursue three extensions. First, we will improve the handling of ambiguous part requests that match multiple components by enabling clarification questions and providing feedback on the system's interpretation. Second, we will evaluate H2H2R in a more complex, end-to-end assembly task to assess the integrated system across a complete collaborative workflow. Finally, we will investigate online monitoring of assembly actions to detect assembly errors as they occur, extending H2H2R beyond preventive task guidance.

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